Kinematic Explicit Solution and Numerical Analysis

Automated Generation System

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1 Explicit Analytical Solution

The planar four-bar linkage can be solved using the Freudenstein equation. By eliminating the coupler angle α_2, we obtain the mathematical model based on three variable coefficients:

$$A(q_1) = 2L_4 \left(L_1 - L_2 \cos\left(q_1\right)\right) \tag{1}$$
$$B(q_1) = -2L_2 L_4 \sin\left(q_1\right) \tag{2}$$
$$C(q_1) = L_1^2 - 2L_1 L_2 \cos\left(q_1\right) + L_2^2 - L_3^2 + L_4^2 \tag{3}$$

Applying the tangent half-angle substitution, the explicit expression for the absolute rocker angle α_3 is derived as:

$$\alpha_3(q_1) = 2\,\mathrm{atan}\left(\frac{2L_2L_4\sin\left(q_1\right) + \sqrt{4L_2^2L_4^2\sin^2\left(q_1\right) + 4L_4^2\left(L_1 - L_2\cos\left(q_1\right)\right)^2 - \left(L_1^2 - 2L_1L_2\cos\left(q_1\right) + L\right.}}{L_1^2 - 2L_1L_2\cos\left(q_1\right) + L_2^2 - L_3^2 + L_4^2 - 2L_4\left(L_1 - L_2\cos\left(q_1\right)\right)}\right) \tag{4}$$

By parameterizing the linkage lengths proportionally such that the scale factor l is algebraically eliminated, we substitute the relative geometry parameters $L_1 = 2, L_2 = \sqrt{41}, L_3 = 5, L_4 = \sqrt{41}$. The completely simplified explicit expression is:

$$\alpha_3(q_1) = 2\,\mathrm{atan}\left(\frac{\sqrt{-\left(-4\sqrt{41}\cos\left(q_1\right) + 61\right)^2 + 164\left(-\sqrt{41}\cos\left(q_1\right) + 2\right)^2 + 6724\sin^2\left(q_1\right)} + 82\sin\left(q_1\right)}{-2\sqrt{41}\left(-\sqrt{41}\cos\left(q_1\right) + 2\right) - 4\sqrt{41}\cos\left(q_1\right) + 61}\right) \tag{5}$$

2 Numerical Simulation

The relative joint angles $\theta_2, \theta_4, \theta_6$ are extracted by compensating for the initial phase offsets at $q_1 = 0°$. The resulting motion curves over a complete 360° driving cycle are plotted in Figure 1.

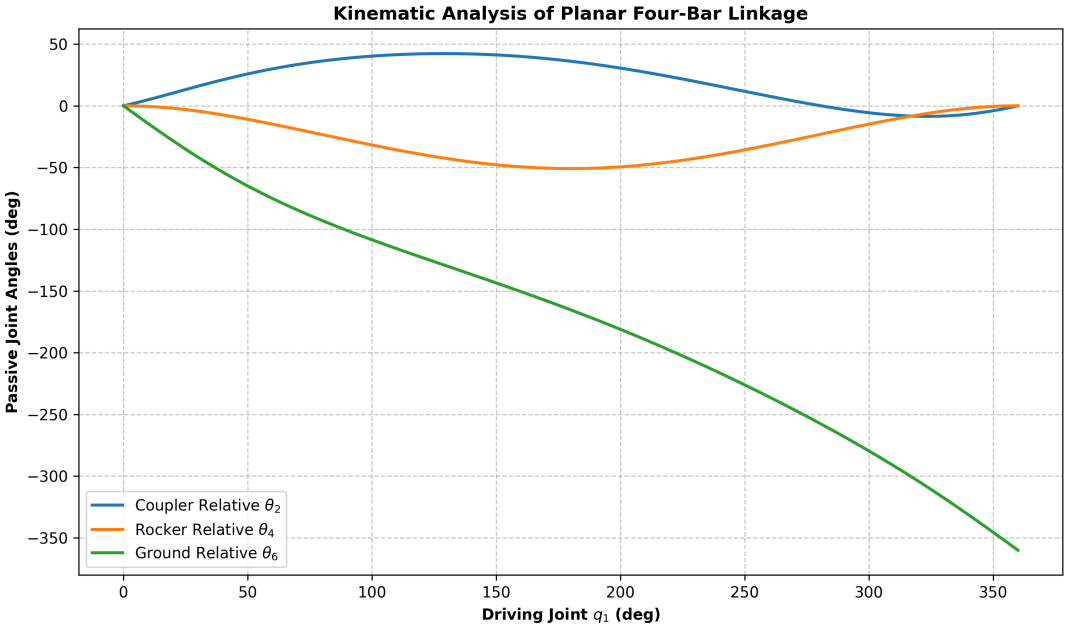


Figure 1: Variation of passive joint angles with respect to the driving joint q_1.